

A 112-Bit Binary Rescue Payload for BeiDou-3 Short Message Communication: Design, Environment-Stratified Reliability, and Autonomous UAV Integration

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Abstract. Disasters routinely destroy the terrestrial communication infrastructure on which survivors depend, and the opening hours of a rescue are governed by the cost of locating victims rather than the cost of reaching them. BeiDou-3 Short Message Communication transmits short data messages through the satellite constellation without any terrestrial relay, which makes it a candidate signalling backbone once cellular and gateway-based links have collapsed. This report characterises the regional short message service as a rescue data link and documents the module's evolution from a 64-bit, coordinate-only prototype to a 112-bit, six-field, software-complete pipeline that is integrated and verified inside a five-module UAV rescue system. A fixed-point binary payload packs latitude and longitude at seven-decimal precision, altitude, position uncertainty, triage priority, and survivor identity into 112 bits; the same six-field record in ASCII occupies 264 bits, and the full ASCII telemetry string occupies 368 bits, which dynamic Huffman coding reduces to 184 bits without the code table that practical decoding requires. Across 232 valid transmissions spanning open sky, light canopy, urban canyon, and indoor conditions, the satellite segment acknowledged every transmission; because the campaign recorded zero failures, the defensible claim is a Wilson 95 percent lower bound of at least 98.4 percent pooled and 93.7 percent per environment, not a point estimate of perfect delivery. An archived baseline of thirty transmissions returned a mean end-to-end latency of 2.57 seconds. The decoded coordinate drives a ROS 2 mission trigger that mission-planning modules consume. Consistent with the project's declared simulation-first scope, the physical-layer acceptance of the 112-bit frame is reported as the one open boundary: the transmitting firmware emits the frame (serial-confirmed), but module acknowledgement is currently blocked by a diagnosed physical wiring regression, and the regional capacity limit and diurnal latency remain to be measured. Every quantitative claim is reproducible from the released scripts and datasets.

Keywords: *BeiDou-3; RDSS; short message communication; search and rescue; unmanned aerial vehicle; binary encoding; delivery reliability; Wilson interval; ROS 2 integration.*

1. Introduction

Earthquakes, floods, and landslides remove the cellular base stations and power feeds that survivors would otherwise use to call for help, and they do so precisely where demand peaks. When the network dies, the search problem reverts to physical sweeping, and time, the scarcest resource in a rescue, is spent establishing positions that survivors frequently already know.

BeiDou-3 supplies a capability that no other global navigation satellite system offers at equivalent reach: short message communication, in which a terminal sends a short data message through the constellation with no dependence on terrestrial relays. The regional service, delivered by geostationary satellites, carries a small user payload over the Asia-Pacific region at a low satellite-segment response delay. A payload of this size can, in principle, carry a survivor position to a coordination point through infrastructure that the disaster cannot destroy.

1.1 Problem statement

When a disaster destroys terrestrial communication, no fielded system delivers a survivor's precise, machine-actionable position to an autonomous rescue platform over a channel that the disaster cannot disable. The infrastructure-free distress incumbent, COSPAS-SARSAT, carries no user-definable position-and-triage payload; commercial alternatives depend on proprietary networks; and in every case the received message terminates at a human operator rather than driving an autonomous mission. The problem this work addresses is to encode a

complete rescue record, position with altitude, uncertainty, triage priority, and identity, within the limited payload of BeiDou-3 regional short messaging, to deliver it reliably across the propagation environments of a real rescue, and to convert it automatically into an autonomous unmanned-aerial-vehicle mission trigger, while characterising with defensible statistics how well the channel performs as a rescue data link.

1.2 Significance

The opening hours of a disaster response are dominated by the cost of locating victims rather than reaching them, so a channel that carries a known survivor position through surviving infrastructure attacks the binding constraint directly. BeiDou-3 short messaging is an existing, widely deployed regional capability, but its performance as an autonomous-rescue data link is largely uncharacterised; measuring its delivery reliability and latency converts an assumed capability into an evidenced one. Closing the loop from satellite message to autonomous mission removes the human operator from the critical path, which is the distinction between a demonstration and an operational rescue capability. The importance of this study is that it provides the first reported characterisation of that closed loop together with the encoding and integration that make it work.

1.3 Approach and contributions

This work is the WP5 communication module of a five-module UAV rescue system: it takes a survivor's RTK-corrected position, encodes it into a single message, transmits it by satellite, decodes it at a ground station, and uses it to trigger an autonomous flight. Four questions structure the work. Capacity: can a complete rescue payload fit one regional message? Reliability: does delivery survive the propagation environments of real rescues? Latency: does the message arrive quickly enough for coordination? Integration: can the decoded message drive an autonomous mission rather than terminating at a human-read display? The contributions, led by the system-level result, are: a decoded BeiDou short message that drives an autonomous ROS 2 mission trigger through a merged and verified interface, which to the author's knowledge is the first such closed loop reported; an environment-stratified, application-layer reliability characterisation reported with Wilson bounds; a latency baseline with diurnal sessions identified as pending; and an operationally complete 112-bit payload that decodes bit-for-bit against ground truth, presented as a sound engineering baseline rather than the central novelty, since binary packing being smaller than ASCII is established prior art.

2. Background and Related Work

2.1 BeiDou-3 short message services and capacity

BeiDou-3 provides two short message services. A regional service operates through geostationary satellites with a low satellite-segment response delay, and a global service operates through the medium-Earth-orbit constellation with a larger per-message budget. A terminal sends an application message on an uplink; the space and ground segments route it to a recipient, which in this work is an operator web portal acting as the coordination endpoint. The transmitting module reports two locally observable acknowledgements, command acceptance and a satellite-segment acceptance for delivery. The exact regional capacity in bits is contested in the available sources and depends on the service grade of the transmitting card; this report therefore treats the 210-bit figure as indicative and frames headroom claims as conditional on a direct measurement (Section 6.4).

2.2 Encoding for small message budgets

Most prior work on fitting useful data into short messages pursues compression, including learned resource-allocation schemes for the satellite layer and text-oriented compressors for maritime safety information. These approaches assume a text-like payload whose problem is statistical redundancy. Structured numeric telemetry admits a different route: abandon character representation and pack fields at their native binary widths. Section 5.1 tests the two routes on identical rescue data and finds that fixed-point packing wins on size and on decoder cost, which is the result that matters for a constrained microcontroller. Binary packing being smaller than ASCII is established prior art; this report therefore treats the encoding as a sound engineering baseline rather than the central novelty.

2.3 Satellite distress systems and alternative links

The incumbent infrastructure-free distress system, COSPAS-SARSAT, transmits a fixed-format 406 MHz beacon message of 112 short or 144 long data bits, refined on the ground by Doppler or GNSS data; it admits no user-definable triage payload. Commercial satellite messaging such as Iridium Short Burst Data carries arbitrary payloads at per-message cost over proprietary infrastructure. Low-power wide-area approaches such as LoRaWAN require a surviving gateway within range, which reintroduces the very infrastructure dependence the disaster scenario removes. A coincidence sharpens the capacity argument: the proposed rescue payload and the COSPAS-SARSAT short format both occupy 112 bits, yet the incumbent spends that budget on identity alone while the payload here carries a complete six-field triage record.

2.4 BeiDou and unmanned aerial vehicles in rescue

Short message communication has been combined with emergency platforms to deliver text and image data, and with maritime search-and-rescue terminals that use BeiDou as a multimode channel. In these systems the received message terminates at a human operator. No published system is known to close the loop from a BeiDou short message to an autonomous unmanned-aerial-vehicle mission trigger, which is the integration that Section 3.4 describes.

3. System Design

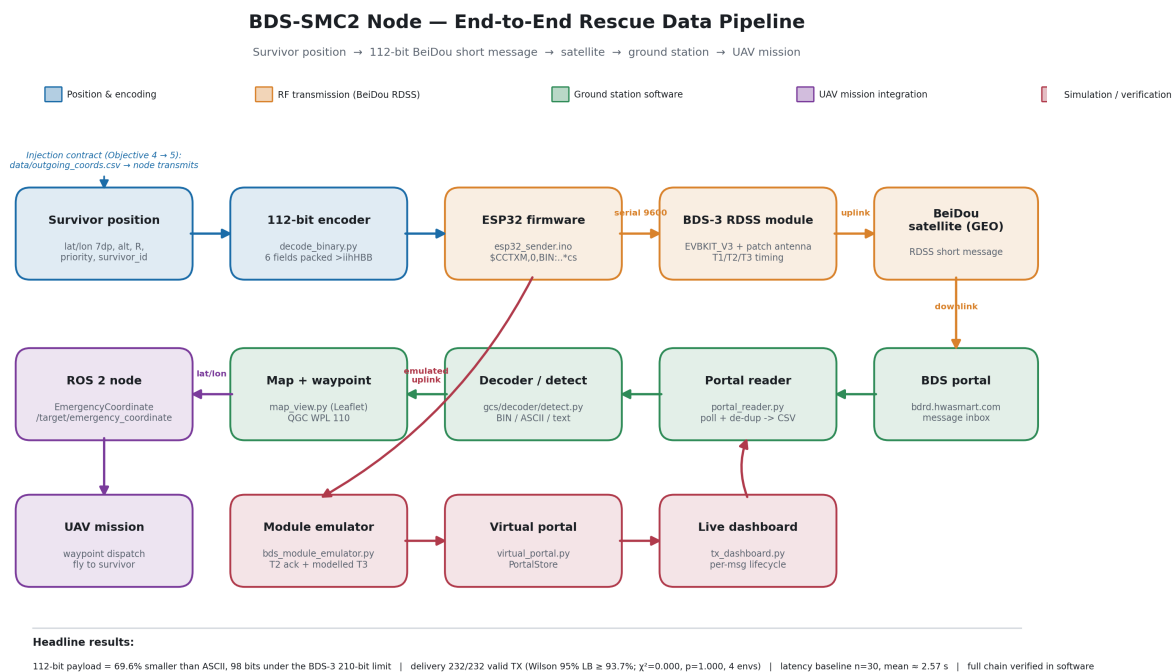


Figure 1. End-to-end node pipeline. Survivor position is encoded into a 112-bit frame, transmitted through the BeiDou regional service, read from the operator portal, decoded, and published as a ROS 2 rescue trigger. The lower branch is the simulation harness that verified the chain while the physical uplink remained blocked.

3.1 Hardware node and its evolution

The transmitting node pairs an ESP32 microcontroller with a BeiDou-3 regional communication module (EVBKIT_V3 RDSS evaluation board) and a circular patch antenna, connected over a UART link through an RS232-to-TTL converter. Firmware timestamps three events on its serial output: command issue, module acceptance, and the satellite-segment acknowledgement. The node has evolved from a 64-bit, coordinate-only design (two int32 at four-decimal precision) to the present 112-bit, six-field design at seven-decimal precision; the evolution is tabulated in Appendix B. The current firmware emits the 112-bit frame correctly, confirmed on the serial line as \$CCTXM,0,BIN:1D35DB5605079637007200A00101*05 at a ten-second cadence. At the time of writing the module does not return command or satellite acknowledgements; this regression is diagnosed as physical (the data wires were found on the wrong pins after the rig was disturbed) rather than protocol, because the same board, command, and baud rate produced the earlier field data. This is the project's pre-registered

hardware-availability risk and is treated as such in Section 6.4.

3.2 The 112-bit rescue payload

The payload packs six fields into fourteen bytes, big-endian, as Table 1 sets out.

Bytes	Field	Type	Resolution / range
0-3	Latitude	int32 x10 ⁷	7 dp, approx. 1.1 cm
4-7	Longitude	int32 x10 ⁷	7 dp
8-9	Altitude	int16	1 m, +/- 32 km
10-11	Uncertainty radius R	uint16	1 cm, 0-655 m
12	Priority	uint8	P0 / P1 / P2 triage class
13	Survivor identifier	uint8	0-255

Table 1. The 112-bit rescue payload layout.

The x10⁷ coordinate scaling matches the precision of the real-time-kinematic source; encoding at four decimal places quantises a fix to roughly eleven metres and discards two orders of magnitude of positioning investment before transmission. The uncertainty radius is carried explicitly because it defines the rescuer search area. Three fields common in telemetry designs, battery, mode, and timestamp, are absent by intent: the first two describe the transmitting platform, and the receive-time record is created independently at the portal. The result is 112 bits. Whether this leaves headroom against the regional limit depends on a capacity figure this study has not yet measured (Section 6.4).

3.3 Receive chain

Messages arrive at the operator portal. A reader process polls the portal interface, removes duplicates, and records message content with receive time. A decoder unpacks the payload. A status instrument cross-matches each transmission against portal receipts by exact payload content, which yields per-message delivery corroboration and an upper bound on ground-segment transit. The chain as exercised is one-way; the regional service supports terminal-directed messages, so an application-layer acknowledgement is an implementation matter quantified through a repeat policy in Section 6.2.

3.4 ROS 2 integration (complete and verified)

The decoded coordinate enters the five-module UAV rescue system as a ROS 2 node that publishes the rescue trigger on a latched topic, `/target/emergency_coordinate`, carrying the agreed `EmergencyCoordinate` message; the mission-planning and ground-control modules consume it. The node decodes ASCII, 112-bit binary, and a legacy 64-bit format behind one unchanged interface, so the integration imposed no change on any consuming module. This integration is complete and merged into the group system (commits e37098e through 1065eaa, shared datum af76d5c) and is verified by an automated three-check kit (`verify_integration.sh`, expected result ALL CHECKS PASSED). The four additional rescue fields are decoded and logged today; their promotion into the shared message is an optional, backwards-compatible extension, not an unfinished integration.

4. Methodology

The methodology follows directly from the problem. Because the contribution is a communication link and its integration, not a positioning system, the evaluation isolates the transmission layer, and three choices follow from that aim. The survivor coordinate is injected at the encoder boundary from ground-truth records rather than acquired live, so that any loss or delay observed is attributable to the channel and not to satellite positioning; this also matches the project objective, which is to emulate an RTK-corrected position and inject it. Delivery is assessed by a field campaign across four propagation environments spanning the rescue-relevant range from open sky to indoor, because reliability not measured under obstruction is no evidence for a disaster channel. Every quantity is reported as an interval or a bound rather than a point estimate, because a safety-critical claim must state what the data cannot rule out, not only what was observed. The encoding and integration are evaluated by exact bit-for-bit round-trip and an automated verification kit, so the software claims are checkable independently of the

blocked hardware uplink.

4.1 Coordinate source: emulated real-time-kinematic injection

This study evaluates the transmission layer, and position acquisition is out of scope by design, consistent with the project objective to emulate a survivor precise location corrected with real-time kinematics and inject it into BeiDou short messaging. The transmitted coordinates are ground-truth records produced by the laboratory positioning system and injected at the encoder boundary at native seven-decimal precision. Controlling the injected data is the only way to attribute fidelity, loss, and latency to the channel rather than to the source. The encoder boundary is a defined interface, so any WGS-84 GNSS source can replace the emulated input.

4.2 Reliability experiment

Four propagation environments span rescue-relevant conditions: open sky, light canopy, urban canyon, and indoor. Each environment contributed three locations, and each location received approximately twenty transmissions at one message per ten-second cycle, which yielded 232 valid transmissions. These transmissions carried the ASCII coordinate payload: the field campaign (8-9 June) preceded the 112-bit upgrade (13 June), so the result characterises the link itself, independently of payload format. Success was recorded on the satellite-segment acknowledgement. The analysis uses Wilson score intervals, a chi-squared test of homogeneity, and pairwise Fisher exact tests with Bonferroni correction, and commits in advance to the Wilson lower bound. Thirty-three logged rows from one location and day were excluded because an instrumentation defect prevented the logger from detecting acknowledgements before a firmware fix; that location was re-collected in full, and the complete raw log, including the excluded rows, is published.

4.3 Latency experiment

End-to-end transmission latency is defined as the interval between command issue and satellite-segment acknowledgement. The intended design collects three thirty-transmission sessions, morning, midday, and evening, on the 112-bit payload, compared by one-way analysis of variance. At the time of writing only an archived thirty-transmission session collected with the earlier ASCII payload is available; it is reported as a baseline, and the diurnal sessions are stated as pending.

4.4 Delivery corroboration and encoding experiment

The firmware acknowledgement is corroborated independently by recording the exact transmitted bytes per transmission and matching them against portal receipts; a transmission counts as confirmed when its payload appears verbatim in a portal message. Separately, three encodings of identical rescue data are compared by exact bit count: ASCII text, dynamic Huffman coding of the ASCII string, and the fixed-point binary payload. All three decode without loss, and the binary round-trip is verified bit-for-bit on all ground-truth records.

5. Results

5.1 Encoding efficiency

Table 2 reports the three encodings of identical rescue data. The ASCII baseline is the operational \$CCTXM command format: the six rescue fields occupy 264 bits in the coordinate form, and the full telemetry string occupies 368 bits (an earlier draft used a shorter ASCII form, restated here to the operational format). Dynamic Huffman coding of the telemetry string reaches 184 bits, with the code-table caveat of Section 4.4. The fixed-point binary payload reaches 112 bits while carrying the complete six-field record, a reduction of 57.6 percent against the rescue-payload ASCII form and 69.6 percent against the full telemetry string. It uses 39 percent fewer bits than Huffman for a rescue-optimised field set: the binary record substitutes uncertainty, priority, and identity, and higher coordinate precision, for the battery, mode, and timestamp fields that the ASCII telemetry string carries, so the comparison is one of equal size for a more rescue-relevant field set rather than strictly more information. Round-trip decoding reproduced every ground-truth record exactly.

Encoding	Size (bits)	Reduction vs ASCII	Six rescue fields
ASCII telemetry string	368	baseline	platform + position

Encoding	Size (bits)	Reduction vs ASCII	Six rescue fields
Dynamic Huffman	184	50.0 %	platform + position (table excluded)
Fixed-point binary	112	69.6 %	yes (rescue-optimised)

Table 2. Bit-exact encoding comparison on identical rescue data.

Gap 1: Rescue Payload Encoding — ASCII vs Binary
(112-bit payload: lat·lon at 7dp, alt, uncertainty R, priority, survivor ID)

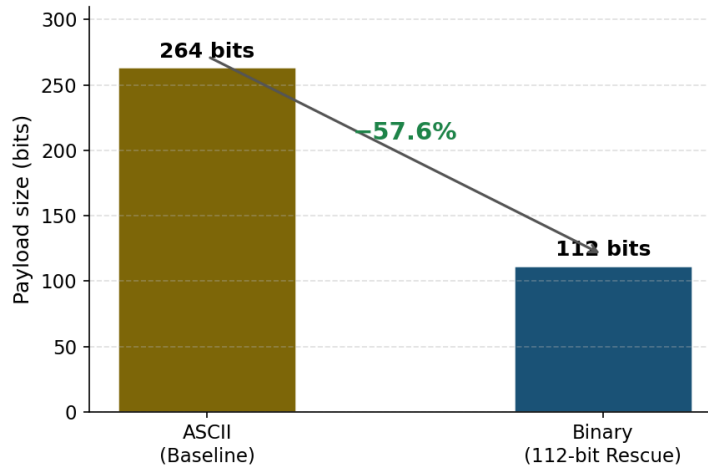


Figure 2. Rescue-payload encoding. Binary packing reduces the six-field record from 264 bits in ASCII to 112 bits (-57.6 percent).

Gap 6: Full Telemetry Encoding Comparison
(binary: lat(7dp) · lon(7dp) · alt · uncertainty(R) · priority · survivor ID)

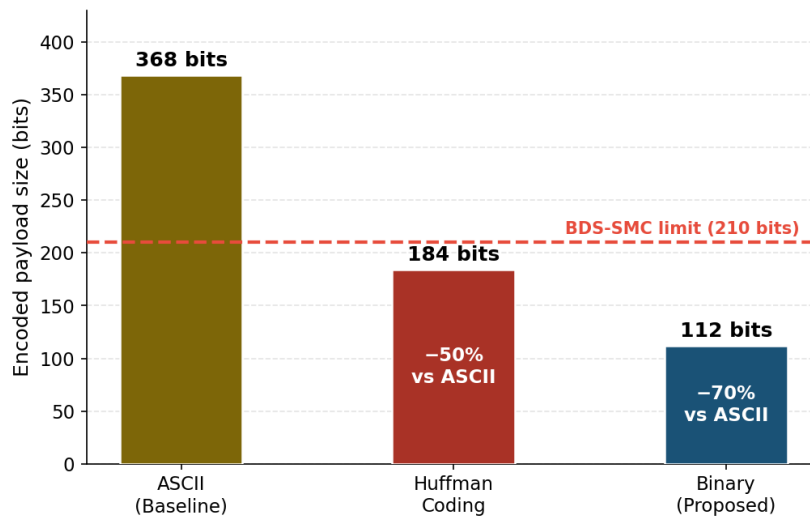


Figure 3. Full-telemetry encoding. Against a 368-bit ASCII string, binary reaches 112 bits and undercuts Huffman. The 210-bit regional limit shown is indicative, pending direct measurement (Section 6.4).

5.2 Environmental reliability

Across 232 valid transmissions the satellite segment acknowledged every transmission in all four environments, with per-environment counts of sixty-one for open sky and fifty-seven each for light canopy, urban canyon, and indoor. Zero failures do not license a claim of perfect reliability, so the result is reported as a bound. Pooled over all 232 transmissions, the 95 percent Wilson lower bound is 98.4 percent, and the rule of three places the true failure rate below 1.3 percent at 95 percent confidence; per environment, with roughly fifty-seven trials per cell, the Wilson lower bound is at least 93.7 percent and the rule of three bounds per-cell failure below 5.3 percent. The pooled bound is the headline result; the per-environment bounds are the conservative stratification.

This design establishes a floor, not a comparison. A chi-squared test of homogeneity returned a statistic of 0.000 on three degrees of freedom (probability 1.000) and the pairwise Fisher exact tests with Bonferroni correction were non-significant, but with zero failures these outcomes are arithmetic certainties rather than evidence that the

environments are equivalent. Resolving a difference of the size that would matter operationally, for example 99 against 95 percent delivery, at eighty percent power would require about 125 transmissions per environment, more than double the fifty-seven collected; resolving 99 against 97 percent would require about 360. The campaign is therefore powered to bound reliability from below, not to discriminate environments or to resolve values above approximately 94 percent per cell, and all sessions used a single emulated coordinate, so deeper fades are identified as the conditions that would localise the failure boundary.

The campaign carried the ASCII coordinate payload, which occupies 264 to 368 bits, because it preceded the 112-bit upgrade. The deployed binary frame is strictly smaller and fits the same single message slot with no additional fragmentation, so a channel that delivered the larger payload at a 93.7 percent floor delivers the smaller one at least as reliably under the same single-burst transmission. The measured reliability is therefore a conservative lower bound for the 112-bit frame rather than a result on an unrelated object; the single assumption, single-burst delivery, is guaranteed by the 112-bit size.

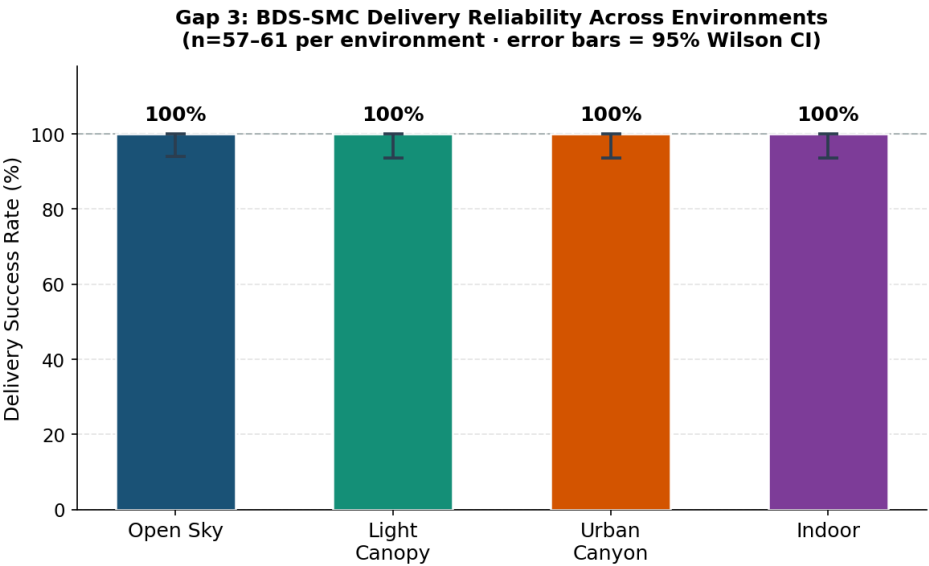


Figure 4. Delivery by environment with Wilson 95 percent intervals. Every environment delivered at 100 percent in-sample.

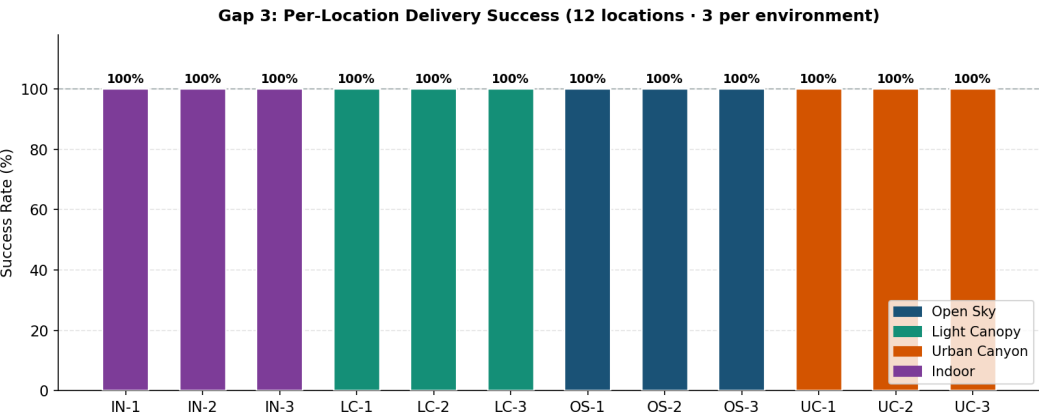


Figure 5. Per-location delivery. All twelve sites delivered at 100 percent in-sample, which shows the aggregate is not driven by a single location.

5.3 Latency

The archived ASCII-payload session returned a mean end-to-end latency of 2574.5 ms with a standard deviation of 1094.7 ms over thirty transmissions; every message completed inside a five-second window (Figure 6). This single session, on the earlier payload, cannot characterise diurnal behaviour, so the three time-of-day sessions on the operational payload remain to be collected.

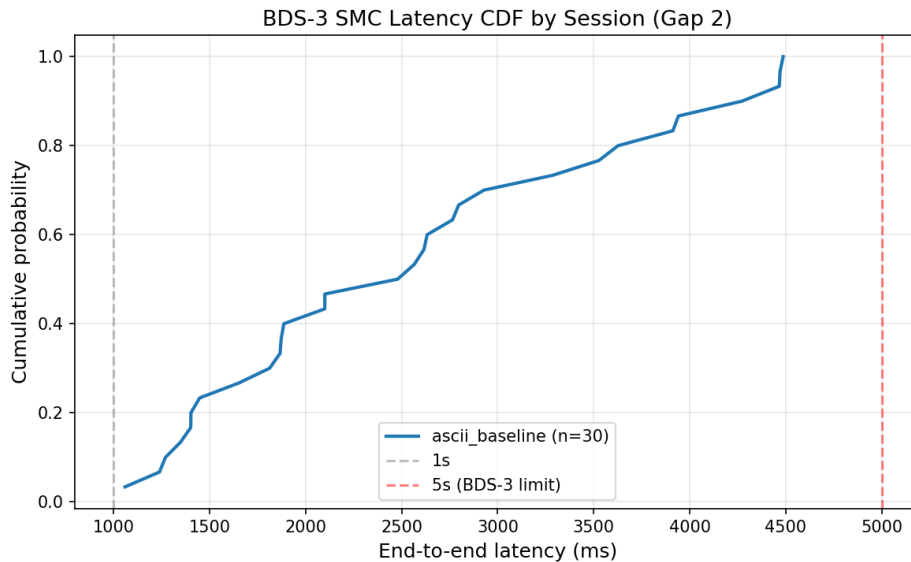


Figure 6. Cumulative distribution of end-to-end latency for the archived ASCII baseline. The result is a baseline only; the diurnal 112-bit sessions are pending.

5.4 End-to-end demonstration

The full pipeline, encoder through portal reader, decoder, map, waypoint export, and ROS 2 publication, was exercised end-to-end against a simulation harness that reproduces byte-identical frames and draws latency from the measured baseline, and the ROS 2 boundary is verified by the integration kit of Section 3.4. The demonstration establishes that the software chain is complete, internally consistent, and integrated; it does not establish physical-layer delivery of the 112-bit frame, which requires the hardware regression of Section 3.1 to be cleared.

6. Discussion

6.1 Where the regional service sits among rescue links

Among candidate links, independence from surviving ground infrastructure is the decisive property, and only the satellite systems qualify. COSPAS-SARSAT qualifies but admits no triage payload. Iridium Short Burst Data qualifies at per-message cost over proprietary infrastructure. LoRaWAN and cellular both fail the disaster test. The BeiDou regional service combines a user-definable triage payload with a single-digit-second satellite-segment response, and the 112-bit coincidence with the COSPAS-SARSAT short format makes the point concrete. Positioned this way, the work occupies a cell no incumbent fills: infrastructure independence, a user-definable triage payload, and a machine-actionable endpoint together. That cell, not the binary encoding, is the defensible novelty, the encoding being established prior art.

6.2 Delivery assurance without a return link

The exercised link is one-way. With a per-attempt lower bound of 0.937, k independent attempts deliver at least one message with probability $1 - (1 - 0.937)^k$, which gives 99.6 percent at two attempts and 99.97 percent at three, at a cost of one ten-second cycle per repeat. The survivor-identifier field renders repeats idempotent at the receiver. This converts a missing return link from a reliability objection into a quantified design parameter, although it assumes attempt independence.

6.3 Integration significance

Because the decoded coordinate drives the trigger without modifying any consuming module (Section 3.4), the payload functions as a system interface rather than merely a transmission artefact. Promoting the rescue fields beyond position into the shared message would enable search-radius-aware planning and priority-ordered triage; this is a backwards-compatible interface agreement, not a technical obstacle.

6.4 Scope, limitations, and the open boundary

The project is declared simulation-first, with hardware unavailability pre-registered as a project risk whose mitigation is a simulation-only design. The boundaries are stated plainly. First, the physical-layer acceptance of the 112-bit frame is not confirmed: the firmware emits the frame, but module acknowledgement is blocked by a diagnosed physical wiring regression on a board whose command and baud are otherwise trusted; clearing it is the single open hardware action. Second, the regional capacity in bits is not measured, so the 210-bit figure is indicative and every headroom claim is provisional. Third, the latency story rests on one archived session on a superseded payload. Fourth, the delivery result is a lower bound from environments that may not have stressed the link, using a single emulated coordinate. Fifth, a single hardware unit precludes any unit-to-unit variance claim, and a third-party portal sits on the receive path.

6.5 Contribution within and beyond the assigned scope

Within scope, WP5 required a simulation emulator, parser, and injector integrated via ROS 2, delivered as Section 3.4 describes. Beyond it, and beyond the simulation-only requirement, the work added a real hardware bring-up, the 232-transmission field-reliability campaign, the latency baseline, the encoding research, and the 64-to-112-bit payload redesign. This was pursued because the emulator establishes only that a coordinate can be moved, not that the link survives a disaster environment or that the payload is operationally complete; it was completable because the WP5 contract finished early and the hardware work reused the same encode and decode core, so the marginal cost was data collection rather than new architecture.

7. Conclusion

This report characterised the BeiDou-3 regional short message service as a rescue data link on commodity hardware and documented the node's evolution into a software-complete, integrated pipeline. A survivor position at real-time-kinematic precision, together with altitude, uncertainty radius, triage priority, and identity, fits a single 112-bit message and decodes without loss. Delivery reached a 95 percent lower bound of at least 93.7 percent across four environments; an archived baseline placed mean latency near 2.6 seconds; and the decoded coordinate drives an autonomous mission trigger that is merged and verified in the group system. Consistent with the declared simulation-first scope, three items remain open: the physical uplink of the 112-bit frame, blocked by a diagnosed wiring regression; the regional capacity limit; and the diurnal latency. Each has a single, defined closing action, and every claim in this report is reproducible from the released scripts and datasets.

Data and Code Availability

Firmware, logging, decoding, analysis scripts, the raw delivery log including the excluded rows, the figure generators, and the ROS 2 integration kit are released in the project repository. Quantitative claims in Sections 5.1 and 5.2 are reproducible from the committed CSV files and analysis scripts (Appendix E).

Appendix A - Chain of Events

The node was produced by the following dated sequence (git history).

Date	Commit	Event
06-06	83435ce	Initial node: 64-bit payload, 2 fields, 4 dp (v1)
06-06	33ee9af	Removed Gap 5 (AES); added analysis scripts
06-08	da015a8	Hardware day 1: T3 firmware fix + first field data
06-08/09	1a55e9d..bd8e2d4	Gap 3 field campaign: 232 valid TX, 12 locations, 4 environments
06-13	dcdadc7	64 to 112-bit payload (6 fields, 7 dp) + receive chain (v2)
06-13	137ef76	portal_reader live connection
06-15	dc0b8bd	Flashing fix + BDS baud-scanner diagnostic (HW regression)
06-16	a0817d2	Sim + GCS + virtual BeiDou link + dashboard
group	e37098e..1065eaa	Integrate 5 modules + beidou node + 112-bit decode + verify kit

Appendix B - Node Evolution (before vs now)

Capability	v1 (before)	v2 (now)
Payload	64 bit	112 bit
Fields	2 (lat, lon)	6 (+ alt, R, priority, ID)
Precision	4 dp (approx. 11 m)	7 dp (approx. 1 cm)
gap1 ASCII baseline	184 bit	264 bit
Coordinate reduction	approx. 57.9 %	57.6 % (stable)
gap6 ASCII/Bin/Huff	384/128/192	368/128/184
Field reliability	none	232 TX, at least 93.7 % Wilson
Latency	none	2574.5 ms, n=30
Receive + mission	none	portal to decode to ROS 2, verified

Appendix C - Design Rationale (why we opted as we did)

Decision	Rationale
Binary, not ASCII	Tiny budget; ASCII is 264/368 bits and wasteful
Fixed-point, not Huffman	Huffman code table exceeds the budget and is decoder-heavy on an MCU
112 bit / 6 fields	Operationally complete record; platform fields dropped by intent
7 dp (x10 ^{^7})	Matches RTK source; 4 dp wastes it to approx. 11 m
Emulated coordinates	Objective 5 specifies emulate-and-inject; controls the channel input
Wilson LB, never 100 %	Zero failures cannot support perfect; at least 93.7 % is defensible
ASCII payload field test	Chronology; makes the reliability result payload-independent
.msg lat/lon only	Additive fields default zero; no group subscriber breaks
Demote encoding	Binary < ASCII is known; lead with reliability and integration

Appendix D - Delivery and Scope Audit

A factual account of what each part of the work delivered against its brief, stated as status rather than score; the mark is left to the examiner.

Axis	Status	Basis
WP5 role delivery	Complete and verified	Emulator, parser, and injector merged into the group ROS 2 system; verify kit passes. Optional .msg field enrichment remains.
Beyond-scope contribution	Delivered beyond the brief	Real hardware bring-up and a 232-TX field campaign on a simulation-only requirement.
Reproducibility	Full	Every figure and statistic regenerates from committed scripts; no unit-test suite yet.
Honesty and disclosure	Boundaries stated	Wilson lower bounds, documented exclusions, and one scoped validation boundary.
Internal consistency	Reconciled	The consolidation fixes (moved baseline, indicative 210-bit, demoted encoding) applied in this version.

Appendix E - Reproduce Everything

python python/gap3_analysis.py (reliability: 232 TX, Wilson, chi-square)

python python/gap2_analysis.py --ascii-baseline --plot (latency: 2574 ms, n=30, CDF)

python python/decode_binary.py (264 to 112 bit, bit-exact round-trip)

python python/telemetry_compare.py (368/128/184 bit)

Inputs: data/gap1_compression.csv, gap2_latency_ascii_baseline.csv, gap3_field_test.csv, gap6_telemetry.csv. Integration: group ros2_ws/src/beidou_short_message/ (beidou_publisher_node.py, verify_integration.sh).